

Nested patterns in urban butterfly species assemblages: respective roles of plot management, park layout and landscape features

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Abstract Increasing numbers of cities are currently developing sustainable policies aimed at promoting urban biodiversity and ecological dynamics through the planning of green networks and the implementation of more sustainable management practices. These human activities can strongly influence environmental factors on which the organization of ecological communities at different scales depends. Thus, it is of fundamental importance to understand the relative impact of local management, green space design and landscape features on the distribution and the abundance of species in urban areas. On the basis of 2 years of butterfly surveys in urban public parks within an extensive Mediterranean metropolitan area, Marseille (South-East France), the aim of this paper is to provide a better understanding of the effect of these three environmental scales (plot, park, landscape) on the composition and organization of species assemblages. Using variation partitioning and nestedness analysis on ecological data aggregated at plot-level and park-level respectively, we demonstrate the preponderant effect of landscape scale features on urban butterfly assemblages. Our results also highlight an important co-variation of plot management, park layout and urban landscape features, in their interaction with the community structure of urban butterflies. Although there is no significant species-area relationship, significantly nested patterns arise in species composition. Selective

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colonization appears as a driving force constraining the constitution of species assemblages within the city. However, a prospective study on adjacent more natural areas suggests that biotic limitations, interspecific competition and habitat filtering may play an important role if a larger portion of the urbanization gradient is explored, which remains to be investigated.

Keywords Rhopalocera · Functional traits · Redundancy analysis · SLOSS · Metacommunity

Introduction

The magnitude and the celerity of the process of urbanization, as a key landscape process affecting biodiversity world-wide (Chiari et al. 2010), requires the identification of general ecological “rules” for refuge design (Cutler 1994), in order to enhance biodiversity in urban areas. Promoting urban biodiversity requires a better understanding of the composition and structure of species assemblages in order to identify ecological processes underlying these patterns. Hence, there is increasing interest among ecologists in remnant nature within and around urban areas (e.g., Bastin and Thomas 1999; Niemelä et al. 2002; Wang et al. 2007; McDonald et al. 2008; Sanford et al. 2009). Many studies also focus on the significance of man-made habitat patches, such as public green spaces, as biodiversity reservoirs in urban areas (Rudd et al. 2002; Cornelis and Hermy 2004; McFrederick and LeBuhn 2006; Smith et al. 2006; Nielsen et al. 2014). How does the management of particular areas in parks affect biodiversity? Does the design of urban parks influence species richness and composition of assemblages, and if so how? To which extent do these local features interact with landscape, and especially the urban matrix?

The aim of the present paper is to address these questions on the basis of the study of butterfly assemblages in urban public parks within an extensive Mediterranean metropolitan area, Marseille (South-East France). Ecological data will be considered at two different levels of information integration: plot level (sampling units) and park level.

First, with data considered at plot level, we will study butterfly-environment relationships to assess the relative influence of three spatial scales (plot, park and landscape) using variation partitioning on four response variables: species richness, total abundance, abundance of multivoltine butterflies and abundance of uni-bivoltine butterflies. Butterflies are known to be suitable model organisms to study rapid environmental changes (Erhardt 1985; Brown 1991). Variations in the structure of species assemblages can be easily linked to habitat and landscape changes induced by human development (Kitahara and Fujii 1994; Blair and Launer 1997; Niell et al. 2007). Decomposing the relative influence of factors related to the three spatial scales on our response variables (Heikkinen et al. 2005; Peres-Neto et al. 2006) is important because human activities, through planning or management practices, can influence environmental factors on which the organization of ecological communities at these three scales depends. More and more cities are currently developing sustainable policies aimed at promoting urban biodiversity and ecological dynamics through the planning of green networks and the implementation of more sustainable management practices (Home et al. 2010). Thus, it is of fundamental importance to understand the relative impact of local management, park design and landscape features on richness and abundance of species in urban areas (Dearborn and Kark 2010).

However, this type of “community-environment” relationship does not take into account specific and/or functional composition. Many butterfly species have specific habitat

requirements (Wiklund 1977), and represent different dispersal abilities and seasonal specificity, as well as various degrees of resource specialisation, allowing a useful classification in various functional groups (Kitahara and Fujii 1997). Moreover butterflies are especially useful when dealing with habitat fragmentation as many species operate as metapopulations (Brückmann et al. 2010). Therefore as a second step, we will study the constitution of species assemblages at park level using nestedness analysis. The concept of nestedness dates from the first half of the 20th century (Hulten 1937), and was first proposed to describe patterns of species composition for insular fauna (Darlington 1957). Nested patterns constitute typical patterns of fragmented landscape where assemblages of species-poor sites are subsets of species assemblages of more diverse sites (Patterson and Atmar 1986; Patterson 1987). Habitat fragmentation is known to be a major feature of urban areas (McKinney 2002), and is expected to generate nestedness in species assemblages as fragmented landscapes are characterized by patches that differ in size and relative isolation (Ulrich 2009; Patterson and Atmar 2000). Initially, two mechanisms were invoked to explain nested patterns among species assemblages: ordered sequences of colonization (Darlington 1957; Simberloff and Wilson 1969) and ordered sequences of extinction (Patterson and Atmar 1986; Patterson 1987). More recently, nestedness analysis has become increasingly popular (e.g., Wright et al. 1998; Fischer and Lindenmayer 2002; Almeida-Neto et al. 2008; Driscoll 2008; Ulrich et al. 2009) as it represents a useful tool to describe patterns of species occurrence, their underlying mechanisms and thus gradients that structure ecological communities (Ulrich 2009). Different processes can in fact lead to nested patterns in community structure: (i) extinction, (ii) colonization, and also (iii) disturbance, (iv) habitat distribution, (v) hierarchical niche relationships and (vi) passive sampling (Patterson and Atmar 2000). These potential causes of nested patterns have been extensively discussed (e.g., Cutler 1994; Driscoll 2008; Ulrich et al. 2009). Thus, nestedness analysis seems to be a suitable tool for understanding gradients structuring the composition of urban species assemblages.

In addition, nested patterns can emerge from metacommunity processes (a metacommunity is a set of local communities connected by dispersal of interacting species) (Wright et al. 1998; Leibold et al. 2004; Logue et al. 2011). In metacommunity models, local species assemblages are seen as the result of interacting local effects (biotic interactions and environmental limitations) and regional effects (dispersal and factors influencing dispersal of species) (Leibold et al. 2004). Both these types of effect are affected by human presence, land-use and activities in urban areas. Four metacommunity models are commonly discussed. In the neutral model, species are considered as functionally equivalent and species assemblages are governed by stochastic factors. In the patch-dynamics model, patches are considered constant in habitat quality and species assemblages depend on tradeoff between dispersal and competitive ability of species. Mass-effect and species sorting models assume that patch quality varies across patches. In the mass-effect model species assemblages are determined by immigration rates, whereas in the species sorting model species composition is primarily determined by habitat filtering (see Leibold et al. 2004 for more details on these models). Here, nestedness analysis can be useful as it helps in distinguishing among the four competing metacommunity models (Driscoll 2008).

Moreover, in conservation ecology, the concept of nested subsets relates to the “Single Large Or Several Small” debate (Boecklen 1997; Patterson and Atmar 2000; Berglund and Jonsson 2003; Ulrich et al. 2009). We will therefore also address the SLOSS question on the basis of butterfly assemblages of urban parks, as this question is fundamental for both theoretical and applied purposes, such as management and planning of urban green spaces.

Material and methods

Study site

This study was carried out in the coastal city of Marseille (France). The region has a Mediterranean climate characterized by cool winters, irregular precipitation (between 700 and 1000 mm per year) and pronounced summer drought that limits the growing season to brief periods in spring and autumn (Joffre et al. 2007).

Marseille covers an area of 240 km² and is the core of the third-largest French metropolitan area with a population estimated at 1,605,000 in 2007. In contrast to many European cities, it is not surrounded by a peripheral agricultural belt, but by calcareous massifs which bring natural and semi-natural areas (i.e., sclerophyllous forests and scrubs) to the gates of the city. This configuration offers a range of landscape contexts from city centre of high building density to natural massifs, and thus different degrees of isolation from regional species source areas.

Fifteen city parks have been selected for study according to size (1 to 15 ha) and surrounding landscape type (see Lizée et al. 2011), so as to be representative of the diversity of urban environments occurring in Marseille (Fig. 1).

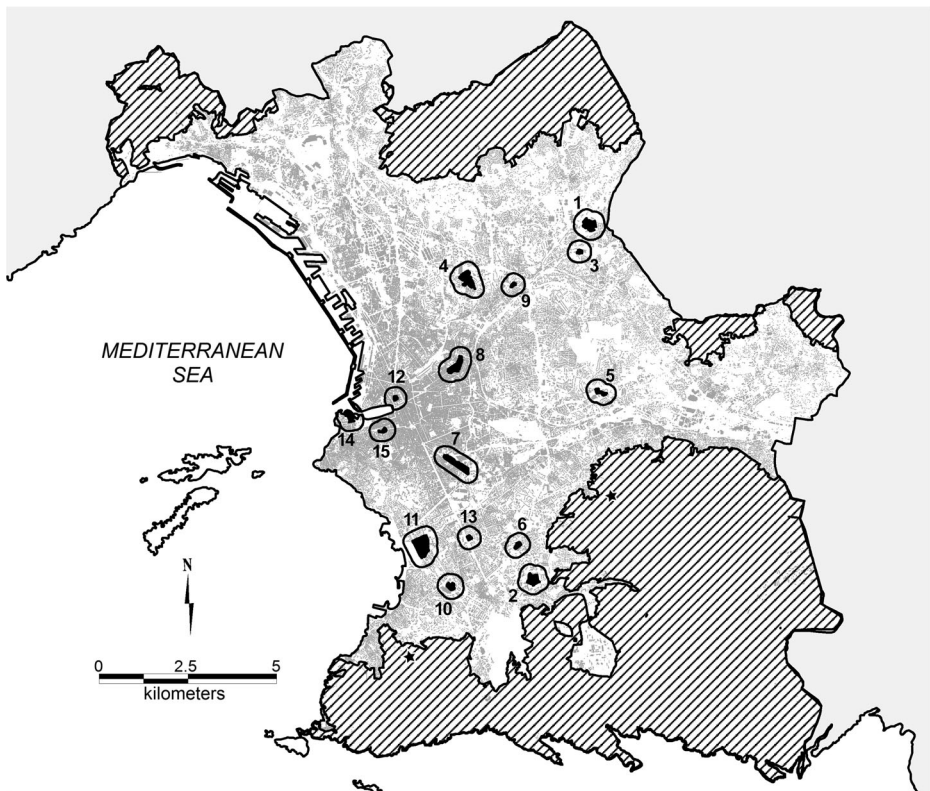


Fig. 1 City of Marseille with buildings (*in grey*), natural areas (hatching), the 15 urban parks studied (*in black*) with their 250 m buffer-zone, and 2 prospective study sites (stars - cf. Discussion). Parks are numbered according to their rank in the Online Resource 3

Butterfly surveys

Butterfly communities were sampled during 4 months in 2008 (May to August) and during 3 months in 2010 (June to August). Seven to 12 transects were performed in each park according to park size, corresponding to a total of 132 sampling plots. Each of these 132 sampling plots has been replicated 7 times during the 2 years of survey (once a month). Transects were chosen in such a way as to represent the habitat diversity of each park. The heterogeneity of the park vegetation led us to a compromise between the usual transect count and point count, in order to collect samples in homogeneous vegetation units. So transects in this study correspond to linear surveys of 25×10 m where all individuals crossing this area were recorded during 5 min. In order to standardise the climatic conditions, plots were sampled between 9.00 and 17.00 on days when the temperature was above 20 °C, cloud cover was less than 50 % and wind speed was less than 20 km/h.

Recorded species were characterised following the functional classification of Kitahara and Fujii (1994, 1997). On the basis of resource utilisation patterns by species, these authors proposed a classification according to 2 fundamental niche dimensions (time and food). Applying the generalist/specialist concept to the analysis of butterfly community structure, they classified butterfly species based on 2 representative traits: larval resource breadth (range of host-plant species used by larvae) and voltinism (number of generations per year). They considered host-plant specificity as an indicator of the niche breadth in a larval food dimension, while the notion of voltinism was extended to serve as a surrogate for the niche breadth in a seasonal time dimension (see Kitahara and Fujii 1994, 1997). This classification resulted in 4 functional groups: (i) typical specialists (TS) which are uni- or bivoltine species that feed on host-plants belonging to 1 plant family only; (ii) typical generalists (TG) which are multivoltine species (3 generations or more per year) that feed on host-plants belonging to 2 or more plant families; (iii) rebound specialists (RS) which are multivoltine species that feed on host-plants belonging to 1 plant family only; and selective generalists (SeG) which are uni- or bivoltine species that feed on host-plants belonging to 2 or more plant families (see Online Resource 1) (Santin 2004; Lafranchis 2000; Tolman and Lewington 1999).

Environmental variables

A total of 11 environmental variables have been used in order to characterize each of the 132 transects. They correspond to three different matrices representing three environmental scales: landscape scale (5 variables), park scale (4 variables) and sampling plot scale (2 variables).

Landscape characteristics

The landscape surrounding each of the 15 parks has been characterized within a 250 m buffer zone. Densities of buildings, density of trees (number of trees per hectare) and density of grasslands have been computed from maps (IGN 2004, 2006). Mean patch area (m^2) has been computed from the cadastral register and distance from park to natural areas surrounding the city was measured from a municipal town planning document.

Park characteristics

For each park, vegetation units have been mapped and habitat types have been characterized using field surveys (Fig. 2). A typology of these vegetation units based on vegetation structure (closed vegetation, open vegetation and intermediate) and intensity of management (high or moderate)

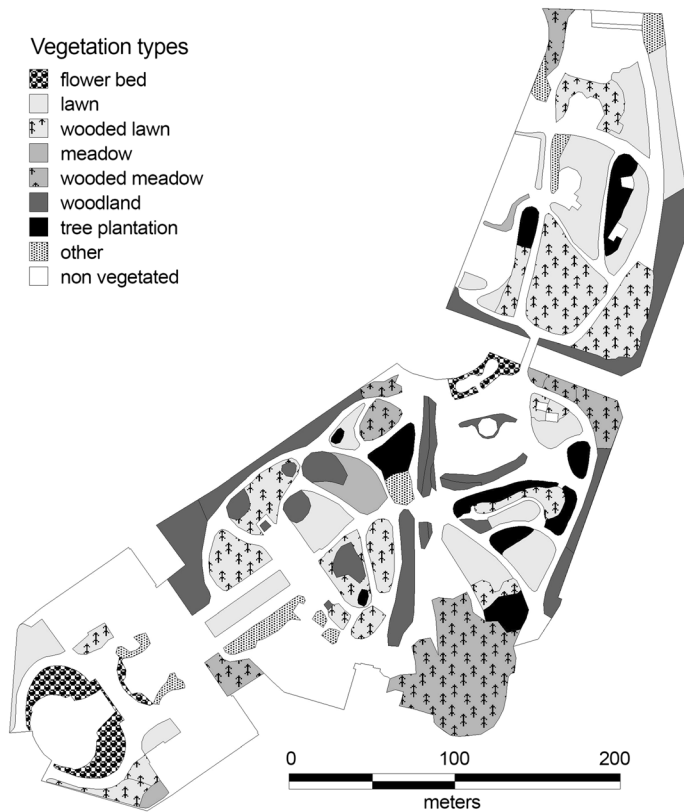


Fig. 2 Example of mapping of vegetation units (Longchamp Park)

resulted in the discrimination of 6 habitat types (lawns, meadows, wooded lawns, wooded meadows, tree plantation, woodland) (cf. Online Resource 2). This classification enabled us to calculate an area-weighted diversity index (Shannon index) for each of the 15 studied parks, based on 8 habitat types (the 6 habitat types previously cited to which were added flower beds, easily identifiable, and the category “other” for vegetation units that did not fit in any of the classes). Park design has also been characterized by total area (hectare), fragmentation (number of vegetation units / total area) and vegetated area (hectare) (difference between the sum of non-vegetated units’ area and total area).

Plot characteristics

For each sampling plot (transect), cover of flowering plants (in percentage of the transect area) and luminous intensity (lux) were recorded at each visit. Means of these 2 variables have been used in order to characterize each sampling plot, reflecting the degree of management and habitat type.

Data analysis

Variation partitioning on butterfly data aggregated at plot level

Here, we aim to assess the relative importance of the three sets of environmental variables on four response variables: species richness, total abundance, abundance of uni-bivoltine

butterflies and abundance of multivoltine butterflies at plot level. As a first step, stepwise selections were performed, within each of the three groups of environmental variables, to exclude variables that did not contribute significantly to the explained variation in the 4 response variables (and thus to avoid saturated global model). Thus, we performed a first series of (partial) regression analyses using redundancy analysis (RDA) and we selected variables within each group on the basis of Akaike information criterion (AIC) (Akaike 1978). We tested, for each environmental variable, both linear and quadratic terms. For each of the three sets of environmental variables (plot, park, landscape), the combination minimising the Akaike criterion was used to select the variables to include in the analysis. Note that for abundance of uni-bivoltine butterflies, analyses were performed on the 93 sampling plots where uni-bivoltine species occurred.

Then, global variation partitioning was performed for each of the 4 response variables in order to evaluate relative influence of the 3 sets of selected variables on our response variables (Borcard et al. 1992; Legendre and Legendre 1998). This procedure allows the quantification (adjusted R squared) of individual and shared fractions of variation in the response variables due to the different matrices of predictor variables. Significance of these different fractions (when testable) was tested using permutation tests (Peres-Neto et al. 2006).

The “Vegan” package (Oksanen et al. 2010) in the statistical software R v.2.9.2 (R Development Core Team 2009) was used to carry out variation partitioning.

Nestedness analysis and the SLOSS question on butterfly data aggregated at park level

Nestedness analysis was performed on a presence-absence matrix with sites in columns and species in rows. The first step in nestedness analysis is the calculation of one or several metrics to quantify the pattern of nestedness in the matrix (Ulrich et al. 2009). Hence, to test whether the species composition of parks exhibited nestedness, we used 3 different metrics: (i) the discrepancy measure, BR (Brualdi and Sanderson 1999); (ii) the nestedness measure based on overlap and decreasing fill, NODF (Almeida-Neto et al. 2008) and (iii) the matrix temperature, T (Atmar and Patterson 1993). Both BR and NODF are interesting metrics as they combine various desired statistical properties: they are invariant to matrix size, shape and fill, they do not treat differentially presences and absences, and they are invariant to matrix transposition (Ulrich 2009). Although, T is positively correlated with matrix size (Ulrich and Gotelli 2007), it was also calculated as it remains to date the most widely used metric to assess nestedness (Atmar and Patterson 1993; Ulrich 2009). Here, we began without any assumption on gradient influencing the degree of nestedness. So the matrix was sorted according to row and column totals (i.e., parks were ordered from the richest to the poorest in terms of species richness, and species were ordered from the most frequent to the rarest). Metrics were then calculated for this packed matrix (see Online Resource 3).

The second step of nestedness analysis is to compare the calculated metrics with appropriate null model or randomization test in order to assess their statistical significance (Ulrich et al. 2009). Thus, to test whether the obtained values of BR, NODF and T differ from what may be expected on a random basis, we compare these values with those obtained from null distributions using the “equiprobable row and fixed column totals constraint model”. This null model has been proved to represent a

satisfying combination of errors of type I (rejecting H_0 when it is true) and errors of type II (failing to reject H_0 when it is false) (Presley et al. 2010). This null model constrains species richness of each site to equal empirical richness with equiprobable occurrences for each species. Thousand permutations were thus performed based on this model to form the null distribution for each of the 3 metrics of nestedness. Note that BR and T, on one hand, and NODF on the other hand, vary in an opposite direction (lower values than expected for BR and T indicate nestedness, whereas higher values than expected for NODF indicate nestedness). As nested subset patterns in community composition may be due to habitat distribution (Cutler 1994; Driscoll 2008), we wanted to test for nestedness in habitat distribution between parks. Thus, we conducted the same nestedness analysis on the basis of the habitat composition of each park (see Online Resource 4).

Both nestedness analyses (for species richness and habitat richness) were performed using the NODF-Program (Almeida-Neto and Ulrich 2011).

Then, to identify gradients of environmental variables that might influence the degree of nestedness, Spearman correlations between ranks of parks in both packed matrices on one hand, and between ranks of parks ordered according to species richness and environmental variables on the other hand, were calculated (Patterson and Atmar 2000). In the same way, Spearman correlation between the degree of rarity of species (species ranks in the packed matrix) and the degree of specialization of species was assessed (see Online Resource 1).

Finally, to test the SLOSS question on our data set, two cumulative species-area curves were constructed: one for the parks ordered from the smallest to the largest and another curve for parks ordered from the largest to the smallest. If the “small-to-large” curve remains above the “large-to-small” curve, this indicates that a set of small sites could harbour more species than a larger park (or a set of larger parks) of the equivalent total area (Berglund and Jonsson 2003; Ulrich et al. 2009). However, if the curves intersect between the extremes, interpreting the importance of patch size is not straightforward (Berglund and Jonsson 2003).

Results

Over the 2 years of the butterfly survey, 2459 individuals of 41 species were recorded within the 15 urban parks studied (see Online Resource 1 for species list). Among these 41 species, 12 TG species (1745 individuals), 5 RS species (268 individuals), 7 SeG species (289 individuals) and 17 TS species (157 individuals) were recorded.

Variation partitioning on butterfly data aggregated at plot level

The results of the first series of partial regressions conducted to exclude variables that did not contribute significantly to the explained variation in the 4 response variables are shown in Table 1.

For all environmental variables, except density of trees and density of grasslands, linear terms were selected by stepwise variables selection. The selection of quadratic terms for the 2 latter environmental variables reveals curvilinear relationships with the 4 response variables.

Scores obtained by the selected environmental variables on the global RDA axis indicate the direction of their relationship with the response variables (species richness, total abundance, abundance of uni-bivoltine butterflies and abundance of multivoltine butterflies)

Table 1 Selected term (*L* linear, *Q* quadratic, *N.S.* not selected) and score on final RDA axis for each of the 11 environmental variables

		Richness		Abundance		Abundance of multivoltine		Abundance of uni-bivoltine	
Plot	Luminous intensity	L	0.41	N.S.	–	N.S.	–	N.S.	–
	Flowering plant cover	L	0.66	L	0.64	L	0.73	L	0.52
Park	Total area	N.S.	–	N.S.	–	N.S.	–	N.S.	–
	Vegetated area	N.S.	–	N.S.	–	N.S.	–	N.S.	–
	Fragmentation	L	–0.51	L	–0.44	L	–0.42	L	–0.52
	Habitat diversity	L	–0.59	L	–0.59	L	–0.57	L	–0.66
Landscape	Density of buildings	L	–0.56	N.S.	–	N.S.	–	L	–0.61
	Density of trees	Q	0.56	Q	0.42	Q	0.41	Q	0.32
	Density of grasslands	Q	0.78	Q	0.88	Q	0.83	Q	0.98
	Mean patch area	N.S.	–	L	0.21	L	0.18	N.S.	–
	Isolation	N.S.	–	N.S.	–	N.S.	–	N.S.	–

(Table 1). Concerning plot characteristics, luminous intensity had a significant effect on species richness only, whereas the presence of resources for adult butterflies (cover of flowering plants) significantly influenced our 4 response variables (Table 1). Selected park characteristics (habitat diversity and fragmentation) had a negative effect on our response variables, while there is no significant relationship between the response variables and park total area or vegetated area, even when these environmental variables are log-transformed (results not shown). Finally, concerning landscape characteristics, it is of interest to note that density of buildings was selected as influent variable on species richness and abundance of uni-bivoltine butterflies only, influencing negatively these 2 response variables. Besides, as expected, density of trees and density of grassland are the most influential variables, and are positively correlated to our 4 responses variables.

In global RDA, our 4 response variables showed high sensitivity to landscape characteristics which brought most of the constraint variation: 54 % for species richness, 60 % for total abundance, 40 % for abundance of multivoltine butterflies and 75 % for uni-bivoltine butterflies (adjusted R^2) (Fig. 3 and Table 2). There was an important part of variation jointly constrained by the 3 levels of environmental variables (see individual fractions [e], [f], [g]), and especially by landscape and park characteristics jointly (see individual fractions [e], and [g]) (Fig. 3).

Finally, it is worth noting that the relative influence of local features (plot characteristics) and landscape features (landscape characteristics) strongly differed between uni-bivoltine and multivoltine butterflies (Fig. 3). Plot characteristics constrained nearly 29 % of the variation in abundance of multivoltine butterflies with a significant individual fraction of 10.4 % (see fraction [a]), whereas these local features constrained only 20 % of the variation in abundance of uni-bivoltine butterflies (which is the lowest adjusted R^2) with a non-significant individual fraction (Table 2).

Nestedness analysis and the SLOSS question on butterfly data aggregated at park level

First, it is important to note that species richness in parks is not correlated to sampling effort (i.e., the number of studied transect) ($R_s=0.41$, $p=0.13$), nor to park area ($R_s=0.34$, $p=0.21$).

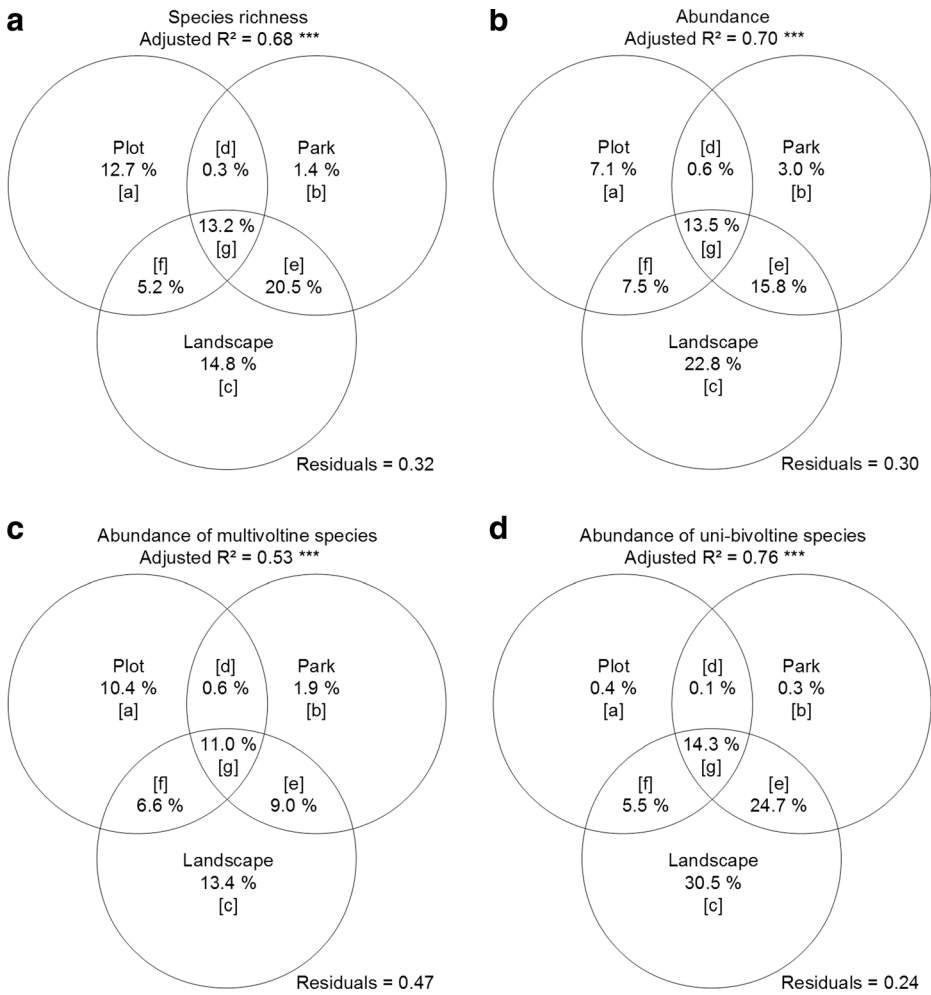


Fig. 3 Results of global variation partitioning for the 4 response variables showing total constraint variation and individual fractions

Table 2 Adjusted R^2 in global RDA for individual and shared fractions and individual fractions alone for each of the three environmental scales

	Richness		Abundance		Abundance of multivoltine		Abundance of uni-bivoltine	
Individual and shared fraction:								
Plot characteristics [a]+[d]+[f]+[g]	0.31	***	0.29	***	0.29	***	0.20	***
Park characteristics [b]+[d]+[e]+[g]	0.35	***	0.33	***	0.22	***	0.39	***
Landscape characteristics [c]+[e]+[f]+[g]	0.54	***	0.60	***	0.40	***	0.75	***
Individual fraction:								
Plot characteristics [a]	0.13	***	0.07	***	0.10	***	<0.01	n.s.
Park characteristics [b]	0.01	*	0.03	***	0.02	*	<0.01	n.s.
Landscape characteristics [c]	0.15	***	0.23	***	0.13	***	0.31	***

The three metrics used to test whether nested patterns occurred in our data set showed significant nestedness in species composition between parks when the matrix was sorted by marginal totals (Table 3). The matrix fill was 40.3 % (see Online Resource 3 for the packed matrix).

Habitat distribution did not show significantly nested patterns between parks (Table 4), except for matrix temperature (T) (Table 4), which is congruent with the matrix fill (72.5 %) as T is known to be sensitive to matrix fill (see Online Resource 4 for the packed matrix). However, ranks of parks between both packed matrices (according to species composition and habitat composition) were not significantly correlated (Spearman $R_s = -0.30$; $p = 0.28$).

Spearman correlations between ranks of parks ordered according to nestedness in species composition highlighted that isolation of parks, composition of the surrounding landscape and fragmentation of vegetation units within parks determined nested structure in specific composition of parks (Table 5). This confirmed the importance of landscape features in structuring butterfly assemblages at plot level (as seen previously), but also at park level. It is also noteworthy that ranks of species in the packed matrix are significantly correlated with the degree of specialization of these species (ranked from 1, typical generalist species, to 4, typical specialist species), indicating that rarest species had a high degree of specialization whereas the most frequent species exhibited generalist traits (Spearman $R_s = -0.70$, $p < 0.001$).

Finally, the analysis of the two cumulative species-area curves showed that, in the set of parks studied, small parks accumulated species faster than large parks of equivalent total area, as the small-to-large curve remained consistently above the large-to-small curve (Fig. 4).

Discussion

Our results demonstrate the importance of landscape features in structuring urban butterfly assemblages when community data are considered at both plot level and park level. The preponderant influence of this broadest scale of environmental variables potentially reveals either dispersal limitation related to dispersal abilities of species and their interaction with urban landscape (Snep et al. 2006; Urban 2004; Ims 1995), or environmental limitation related to human disturbance of urban landscape and/or habitats (Ulrich et al. 2009; Laurance 2008).

Relative importance of plot, park and landscape features

First, variation partitioning showed an important part of variation of each response variable constrained jointly by the 3 environmental scales. This indicates a strong co-variation of plot management, park design and urban landscape features, especially in their interaction with community structure of urban butterflies (here, species richness, total abundance, abundance of multivoltine butterflies and abundance of uni-bivoltine butterflies). Moreover, the main part of

Table 3 Results of nestedness analysis of species composition showing observed and simulated values of BR, NODF and T with 95 % confidence interval and associated probabilities

Metric	Matrix value	Simulated value	L 95 % CL	U 95 % CL	P (H0)
BR	50.00	105.90	99.00	112.00	0.001
NODF	64.63	44.64	40.89	48.01	0.001
Temp	19.22	52.29	46.80	58.05	0.001

Table 4 Results of nestedness analysis of habitat composition showing observed and simulated values of BR, NODF and T with 95 % confidence interval and associated probabilities

Metric	Matrix value	Simulated value	L 95 % CL	U 95 % CL	P (H0)
BR	14.00	16.17	13.00	19.00	0.119
NODF	71.30	67.59	63.86	71.50	0.056
Temp	14.93	35.27	21.41	47.76	0.001

variation in community structure constrained by park characteristics is shared with landscape characteristics, contrasting with the pure effect of park features which does not exceed 3 % (for total abundance). As total area and vegetated area of park are not correlated with our response variables, park characteristics are reduced to diversity and fragmentation of habitat units which are both negatively correlated with the response variables. Here, the negative relationship between butterfly richness (or abundance) and habitat diversity is quite surprising. Indeed, this relationship is known to be positive and monotonic (Hortal et al. 2009). Moreover, in their review of 24 empirical data sets, Hortal et al. (2009) showed that the relationship between species richness and habitat diversity is never negative (at least in natural areas). It seems that habitat diversity measured in our study reflects quite the same organisational pattern as fragmentation. The more the park contains vegetation units, the more the park is diversified in terms of habitats, but also the more the vegetated area is fragmented. This assumption is congruent with the absence of significant species-area relationship (Lomolino 2000; Triantis et al. 2006). Multiplying the different vegetation units in a park, may reduce the size of these units, by increasing the “edge effect”, making them too small to be useful as habitats, and at the same time, contributing to reduce species richness. However, the first outcome of this study is to highlight the need for further investigation, and especially interdisciplinary studies, in order to understand these links between urbanization context (landscape characteristics) and park design, or even management practices (plot characteristics). To what extent do landscape and human components, such as management or use of urban green spaces, interact in urban areas (Cadenasso et al. 2006)?

Although plot characteristics share an important part of constrained variation with landscape and park scales, the pure effect of these local features is greater than the effect of park features. Here, plot characteristics correspond to cover of flowering plants, except for species richness which is also affected by luminous intensity. This strong influence of resource variables on richness and abundance of butterflies has already been demonstrated (e.g., Loertscher et al. 1995; Heikkinen et al. 2005; Thomas et al. 2001; Matteson and Langelotto 2010). Presence of adult butterflies on a plot will depend on the presence of host-plants for

Table 5 Significant correlations (Spearman Rs and associated probabilities of being uncorrelated) between ranks of parks in the matrix packed according to species richness and environmental variables

	Environmental variables	Spearman rs	p (uncorr)
Park rank (nestedness in species composition)	Fragmentation of park	0.64	≤0.01
	Density of buildings	0.65	≤0.01
	Density of trees	−0.64	≤0.01
	Density of grasslands	−0.70	≤0.01
	Isolation	0.54	<0.05

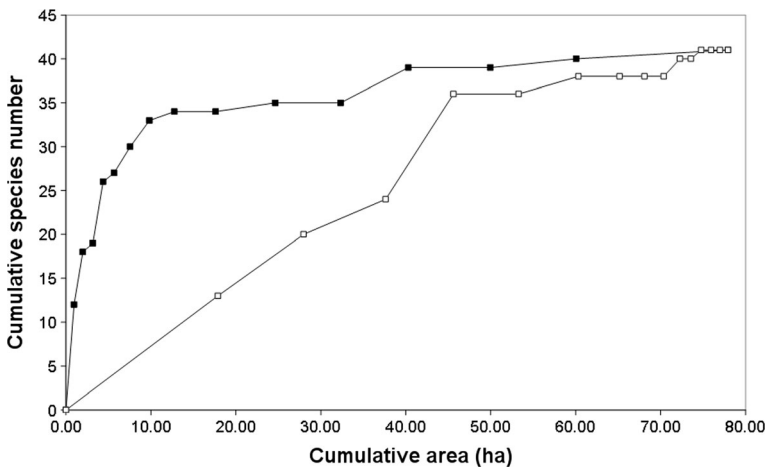


Fig. 4 Cumulative species-area curves. Filled squares represent the situation when the 15 parks were ranked from small to large. Open squares represents the situation when the 15 parks were ranked from large to small

oviposition, but also of nectaring resources which sometimes may even drive adult behaviour and patch occupancy. Some species are indeed more strongly influenced by the search for nectar plants rather than oviposition sites, especially income-breeding species that rely on adult foraging during their reproduction (Janz 2005; Janz et al. 2005; Wiklund et al. 2001; Tammara and Haukioja. 1996; Murphy et al. 1984).

However, the most striking pattern is the difference in the response of uni-bivoltine species and multivoltine species to landscape and local features. The pure effect of plot characteristics on abundance of uni-bivoltine butterflies is not significant, whereas these local features constrain a non-negligible and significant part of variation in abundance of multivoltine butterflies. In contrast, abundance of uni-bivoltine butterflies depends essentially on landscape characteristics which constrain 75 % of the variation, whereas abundance of multivoltine butterflies is the least impacted response variable accordingly to landscape characteristics. This difference in responses of species according to voltinism is interesting because this trait reflects, in part, the length of adult flight period and the time of adult appearance (what Kitahara and Fujii (1997) consider as “seasonal specificity”). Although the relationship between voltinism and “seasonal specificity” is imperfect, this trait may nevertheless be interpreted in terms of dispersal opportunities, and so ability to disperse among habitat patches and to find resources within the urban landscape (Ericson et al. 1988; Fischer and Lindenmayer 2007). Thus, if voltinism is seen as a proxy for dispersal opportunities (as multivoltine butterflies benefit from several generations to disperse through the urban landscape), our results are congruent with several studies related to species colonization and dispersal in fragmented landscape (e.g., Stevens et al. 2010; Snep et al. 2006; Heger and Trepl 2003). The effect of landscape characteristics should be greater for poor dispersers than for good dispersers which will be more likely to maintain viable populations (Fischer and Lindenmayer 2007; Ims 1995). Thus, we can hypothesize that multivoltine species are less strongly affected by landscape characteristics because of their greater opportunities for finding suitable habitats (Campbell 2010). Then, the major influence of landscape feature on abundance of uni-bivoltine butterflies could be linked to dispersal limitations (Urban 2004; Croci et al. 2008). In addition, the influence of density of buildings, that negatively affects abundance of uni-bivoltine butterflies and species richness, is consistent with this assumption. Inversely,

multivoltine species are much more dependent on local features as maintenance, after successful dispersal and colonization, depends on within-habitat features (Crocì et al. 2008). In fragmented landscapes, such as urban areas, resources (when present) could be less easily accessible to less mobile species (Lundholm 2006; Pickett et al. 2001), which may explain the lack of variation in abundance of uni-bivoltine species constrained by the pure effect of local features.

Moreover, this functional trait is also strongly correlated with a range of other ecological attributes (Barbaro and van Halder 2009; Laurance 1991). First, voltinism corresponds to reproductive parameters and breeding phenology (Barbaro and van Halder 2009), but is also associated with feeding habits. In our study, most of multivoltine species are polyphagous (of the 17 multivoltine species, 12 are polyphagous), while most of uni-bivoltine species are oligophagous (of the 24 uni-bivoltine species, 17 are oligophagous) (see Online Resource 1). Hence, multivoltine species can benefit from particular management practices in urban parks (such as flower planting and water supply), which favour the maintenance of exploitable resources during the period of summer drought that “seasonal specialist” species may not be able to exploit in one or two generations (Blair and Launer 1997; Graves and Shapiro 2003). Indeed, typical Mediterranean species, mostly uni-bivoltine, are often diapausing during this summer drought period. Thus, multivoltine butterflies could correspond to performing “urban-adapters” (Blair and Launer 1997), especially under the Mediterranean climate, able to exploit urban resources better than Mediterranean species.

Gradients structuring nested patterns of urban butterfly communities

Now that we have studied relationships between environmental variables and structure of butterfly assemblages at plot level, the question of specific and functional composition remains open. The preponderant influence of landscape characteristics on community variables at plot level, structuring urban butterfly assemblages, is congruent with the results of nestedness analysis which demonstrates that nested patterns are structured along a gradient of landscape variables: density of buildings, of trees, of grasslands and isolation of parks. Hence the first interesting result in nestedness analysis is that nested patterns are not related to park area, but to landscape characteristics summarizing the urbanization gradient.

Ulrich (2009) proposed in his review on nestedness analysis different mechanisms which can cause nested patterns in species occurrence matrices. He distinguishes various hypotheses corresponding to gradients in site properties and species properties: passive sampling, selective colonization, selective extinction, nested habitat and selective environmental tolerance (See Ulrich 2009 and Ulrich et al. 2009 for details on these gradients). To these potential causes of nestedness, we can add the four theoretical metacommunity paradigms (e.g., Leibold et al. 2004), which are processes that can, in some cases, generate nestedness in species composition (Driscoll 2008): neutral model, patch-dynamics model, mass-effect model and species-sorting model.

First, nestedness in our study cannot correspond to passive sampling as species richness (i.e., ranks of parks in the packed matrix) is not correlated to sampling effort (Cutler 1994; Ulrich et al. 2009). As park area is not correlated to nestedness, passive sampling (again) and selective extinction can be questioned, as both these hypotheses predict a positive relationship between species richness and patch area (Ulrich et al. 2009; Patterson and Atmar 1986, 2000, for selective extinction; Cutler 1994 for passive sampling). However, it is difficult to settle the question of potential selective extinction. In a given set of species assemblages, if nestedness is correlated with patch area but not with isolation, the system should be extinction-driven

(Bruun and Moen 2003). But the opposite situation (which is our case) is less clear because either immigration or extinction can generate nestedness according to isolation (Ulrich et al. 2009). Data on species present in urban green spaces in Marseille before the “post-war demographic boom” period could help to resolve this question of selective extinction.

Another common cause of nested patterns in species composition is habitat nested patterns (Ulrich 2009; Cutler 1994; Driscoll 2008). But in our study, nestedness in species composition between parks is not related to nestedness in habitat distribution. However, this nestedness in species composition may be due to nested habitat quality, if disturbance (here, anthropogenic disturbance) is seen as one aspect of habitat (Ulrich et al. 2009). Thus, disturbance can generate nested patterns in species composition due to differences in disturbance susceptibility of species (i.e., selective environmental tolerance) (Worthen et al. 1998; Fernández-Juricic 2002; Ulrich et al. 2009). Disturbance is known to be an important predictor of species composition because adaptations that promote environmental tolerance often trade off against adaptations that promote fitness in situations of intense interspecific competition (Urban 2004).

Thus, taking into account the four theoretical metacommunity models, we can first exclude the neutral model (Hubbell 2001) because, in our study, species are filtered according to functional traits (cf. species ranks in the packed matrix – Online Resource 3) and are obviously not functionally equivalent. The patch-dynamics model (Levins and Culver 1971) can also be excluded as parks (differing in size, design and locality) cannot be considered constant in terms of habitat quality whereas patches are assumed to be identical in this theoretical model (Leibold et al. 2004). Therefore, if disturbance generates nested habitat patterns, this may lead to nested patterns in species composition predicted by the species sorting model (Leibold 1998) or the mass-effect model (Mouquet and Loreau 2003) of metacommunity. In effect, nested habitats can produce nested patterns in species assemblages under both these theoretical models (Driscoll 2008).

As filtering of species in our study indicates that species vary in their dispersal opportunities, seasonal specificity as well as in feeding specialization, the mass-effect paradigm may apply to good dispersers able to cope with highly modified urban habitats (low disturbance susceptibility), but the species-sorting paradigm to those with low dispersal abilities and high habitat specificity (high disturbance susceptibility) (Leibold et al. 2004; Schleicher 2010). This assumption is consistent with the species composition found in adjacent natural areas, consisting in typical Mediterranean sclerophyllous forests and scrubs (Table 6). In these natural areas, dominant species are Mediterranean TS species, whereas TG species, dominating urban public parks, are represented by very few individuals. This may be due to the fact that metacommunity paradigms are not always distinct and especially mass-effect and species sorting models (Logue et al. 2011). With low dispersal, the mass-effect model indeed approaches a species sorting process (Driscoll 2008). In this study, we have focused our analysis on urban parks which constitute only a portion of a larger urbanization gradient. Dispersal limitation and environmental limitation (related to disturbance) seem to be preponderant in this urban portion, but the dominance of Mediterranean and/or specialist species in adjacent natural areas gives ground for considering an important role for interspecific competition and habitat filtering in less disturbed habitats according to the species sorting model (Leibold 1998).

Conservation value of urban parks

Considering the SLOSS question, only perfect nested subsets combined with a significant species-area relationship should give rise to the situation where a single large site harbours more species than any combination of several small sites of equivalent total area (Boecklen

Table 6 Comparison of the most abundant species (with functional classification) in urban public parks and peripheral calcareous massifs during the summer 2008 (Mediterranean species are marked with *)

	Larval diet	Volitinism	Functional group	% of individual in urban parks	% of individual in calcareous massifs
Most abundant species in urban parks in 2008	<i>Pieris rapae</i> (Linnaeus, 1758)	multivoltine	TG	0.28	0.01
	<i>Pararge aegeria</i> (Linnaeus, 1758)	multivoltine	TG	0.10	absent
	<i>Polymniatus icarus</i> (Rottenburg, 1775)	multivoltine	TG	0.10	<0.01
	<i>Maniola jurtina</i> (Linnaeus, 1758)	univoltine	SeG	0.07	0.04
	<i>Colias crocea</i> (Geoffroy, 1785)	multivoltine	RS	0.06	absent
	<i>Arctia agestis</i> (Denis & Schiffmüller, 1775)	multivoltine	TG	0.04	absent
Most abundant species in calcareous massifs in 2008	<i>Coenonympha dorus</i> (Esper, 1782)	univoltine	TS	<0.01	0.24
	<i>Pyronia bathseba</i> (Fabricius, 1793)	univoltine	TS	<0.01	0.17
	<i>Gonopteryx cleopatra</i> (Linnaeus, 1767)	univoltine	TS	0.01	0.12
	<i>Lasiommata megera</i> (Linnaeus, 1767)	multivoltine	RS	0.03	0.12
	<i>Limnitis reducta</i> (Staudinger, 1901)	bivoltine	TS	<0.01	0.05
	<i>Satyrus exul</i> (Hübner, 1804)	univoltine	TS	<0.01	0.05

1997; Fischer and Lindenmayer 2005; Ulrich et al. 2009). But perfectly nested patterns are rare (Fischer and Lindenmayer 2005), and our study showed an absence of significant species-area relationship, demonstrating that small parks accumulated species faster than large parks of equivalent total area. Thus, depending on specific similarity, several small sites can harbour more total species than a larger one. In addition, as landscape characteristics highly influence community structure in the set of parks studied, small parks may accumulate more species because they benefit from subsidies from larger habitat patches in their surroundings (cf. “subsidized island hypothesis” – Anderson and Wait 2001). Moreover, the larger the parks, the more diversified they are in habitat types and the more fragmented they are, which negatively influences species richness. This may also explain why smaller parks accumulate species faster than large parks. Hence, our study highlights that this link between landscape context and characteristics of parks in terms of design and management merits further study, and especially collaboration between natural, social and human sciences.

Conclusion

Finally, the nestedness and metacommunity frameworks represent relevant theoretical models to investigate variations in community structure. We have shown the preponderant influence of landscape features on the structure and composition of urban butterfly assemblages at both plot level and park level. This result calls for further investigation on urban landscape, which seems to constrain many environmental factors and processes structuring the metacommunity. Actually, many questions remain concerning the role of the urban matrix (Lizée et al. 2012), and especially how the characteristics of this matrix may alter the species-area relationship.

Selective colonization has appeared as a driving force constraining the constitution of species assemblages within the city. But the prospective study on adjacent more natural areas (Table 6) suggests that biotic limitations, interspecific competition and habitat filtering may play an important role if a larger portion of the urbanization gradient is explored. However, biological interactions, dispersal and environmental filtering are not exclusive, but rather complementary mechanisms (Logue et al. 2011). The next step is now to understand and quantify their relative importance, which remains to be investigated. Data on differential rates of species dispersal are lacking and would help to clarify the question.

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